








Modelling Financial Markets

Chapter 19: Review and Project Presentations



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What Turns a Result into a Finding?

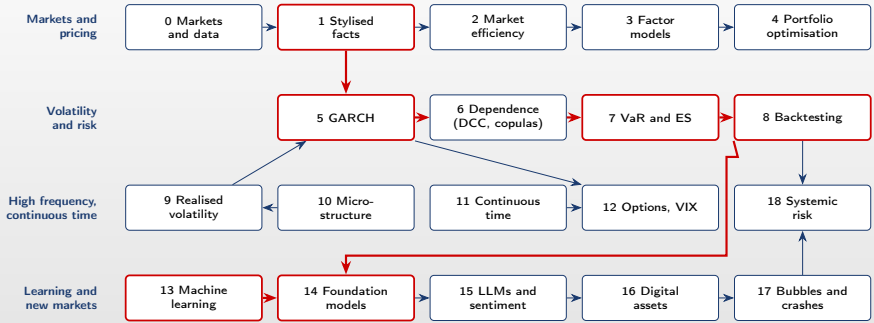
The Chapter in Brief

- The question: when does an empirical result about markets deserve trust, and when does it only reflect the data, the design or luck?
- Part I: the course in one picture, a review of Chapters 0–18, six pitfalls, and one integrated case study, from raw prices of the BET (the benchmark index of the Bucharest Stock Exchange, BVB) to a backtested value-at-risk (VaR) at 1% (chapter summaries with their formulas: ▸ Appendix)
- The main thread: GARCH (generalised autoregressive conditional heteroskedasticity) forecasts volatility, DCC (dynamic conditional correlation) the correlations, VaR and ES (expected shortfall, the mean loss beyond VaR) turn them into a loss
- Part II: the team project, artificial intelligence (AI) in the project, a research case study on nonstandard errors, one pass of the AI discovery loop

What you should be able to do

- Take one question from data to a backtested risk forecast with valid inference
- Recognise the six pitfalls in your own code
- Write a project plan with replication, identification, power and a specification grid

The Course in One Picture



What Connects the Steps of the Analysis

- Red: the main thread, from stylised facts (Chapter 1: heavy tails, volatility clustering, almost no linear predictability) to risk forecasts we can test (Chapters 5–8: GARCH gives $\sigma_{t+\mathbf{1}}$, VaR and ES the loss, backtesting says whether to trust it)
- Chapters 9–12 measure volatility better (intraday data, option prices, the VIX implied-volatility index of the S&P 500) and in continuous time
  - Chapters 13–15 bring flexible models, including large language models (LLMs), always judged with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system
- Special Chapter 20: realised volatility with path signatures on the VOLARE data (VOLatility Archive for Realized Estimates)

Five Questions for Every Model (1/2)

Before estimating

- **Data**: which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model**: which stylised fact does it capture?
- **Model**: which does it ignore?
- **Estimation**: on which window, with which information at time t?

Five Questions for Every Model (2/2)

After estimating

- **Evaluation**
  - Out of sample, with a loss that matches the target?
  - QLIKE (the quasi-likelihood loss) for variance forecasts, FZ0 (the Fissler–Ziegel score) for a (VaR, ES) pair
  - The right diagnostic tests (e.g. Kupiec)?
- **Uncertainty**: robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

Chapter 0: Financial Markets in 2026

- Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
- 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year (CAGR) with 17.6% volatility; Bitcoin 60.6% a year (CAGR) with 66.7%
- BET, July 2007 to February 2009: a drawdown (the fall from the previous peak to the trough) of −82.5%

Chapter 1: Stylised Facts and Returns

- Returns are uncorrelated but not independent: their magnitude is predictable, but their sign is largely unpredictable
- Tail index α: the probability of a loss above x decays like $x^{-\alpha}$; the Hill estimator uses the k largest losses, ordered $L_{(1)} \geq \cdots \geq L_{(k+1)}$ Hill (1975)

$$\hat{\alpha} = \Big[\frac{1}{k}\sum_{i=1}^{k} \ln \frac{L_{(i)}}{L_{(k+1)}}\Big]^{-1}, \qquad \mathrm{SE}(\hat{\alpha}) \approx \frac{\hat{\alpha}}{\sqrt{k}} \text{ for i.i.d. losses}$$

- BET, $k = 153$: $\hat{\alpha} = 2.39$, block-bootstrap 95% confidence interval (CI) $[2.12, 3.03]$
- The i.i.d. interval $\hat{\alpha} \pm 1.96\,\hat{\alpha}/\sqrt{k} = [2.01, 2.77]$ is narrower: large losses come in clusters
- $\alpha < 4$: the fourth moment is likely infinite
- Days beyond 5σ are more than 1000 times more frequent than the Normal distribution predicts (under the Normal distribution: $2\Phi(-5) \approx 5.7 \times 10^{-7}$ a day, one such day in about 7 000 years)

Chapter 2: Market Efficiency

- The efficient market hypothesis: prices fully reflect the information set
- A test of abnormal-return predictability is a joint test of efficiency and the expected-return model Fama (1970)
- Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- Statistical rejection is not exploitable profit

Chapter 3: Factor Models and Asset Pricing

- Expected returns should reward only exposure to systematic factors
- 25 size and book-to-market portfolios: SML (security market line) slope -4.4%
- The GRS statistic (joint test that all intercepts are zero) is 4.18, far above the 5% critical value of $F(25, T - 26)$, below 1.6 on this sample: the CAPM (capital asset pricing model) is rejected ($p < 0.01$)
- Sector ETFs: the first principal component (the market) explains 66% of the variance

Chapter 4: Portfolio Optimisation

- Markowitz asks the right question, but estimation error dominates the answer Markowitz (1952)
- 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
- 96% of the estimated tangency portfolios are truly worse than 1/N

Chapter 5: Conditional Volatility: GARCH

- Volatility clusters, so today's shock predicts tomorrow's variance Engle (1982), Bollerslev (1986)
- The GARCH(1,1) model and its h-day forecast (ε_t: the shock; α, β: the weights of the last shock and of the last variance)

$$\sigma^2_{t+1} = \omega + \alpha\,\varepsilon_t^2 + \beta\,\sigma_t^2, \qquad \bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$$

$$\sigma^2_{T+h\,|\,T} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}\left(\sigma^2_{T+1\,|\,T} - \bar{\sigma}^2\right)$$

- The persistence $\alpha+\beta$ is the rate at which the forecast returns to the long-run level $\bar{\sigma}^2$; the half-life is $h_{1/2} = \ln 0.5 / \ln(\alpha+\beta)$
- S&P 500, 2000–2026: $\alpha+\beta = 0.9947$, so $h_{1/2} = 131$ days
- BET: profile-likelihood 95% CI for $\alpha+\beta$ is $[0.975, 1]$ (IGARCH, the integrated GARCH with $\alpha+\beta=1$, is not rejected): the half-life is at least 27 days, with no finite upper bound

Chapter 6: Multivariate Volatility and Dependence

- Correlations move with the macro regime; diversification is not guaranteed
- SPY–TLT (long US Treasury bond ETF): ρ from -0.40 to 0.11 after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
- Bitcoin–SPY DCC correlation: 0.04 before 2020, 0.31 since

Chapter 7: Value-at-Risk and Expected Shortfall

- VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
- Definitions, with q_α the α-quantile of the returns r (continuous distribution)

$$\mathrm{VaR}_\alpha = -q_\alpha, \qquad \mathrm{ES}_\alpha = -\mathbb{E}\big[r \mid r \le q_\alpha\big] = \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_u \, du$$

- S&P 500, daily: historical $\mathrm{VaR}_{1\%} = 3.45\%$, $\mathrm{ES}_{1\%} = 5.01\%$
- The Normal distribution gives only 2.80%: heavy tails matter

Chapter 8: Backtesting and Evaluating Risk Forecasts

- Test the number and the timing of breaches: coverage and independence Kupiec (1995), Christoffersen (1998)
- Kupiec, with x breaches in T days and the observed rate $\hat{\pi} = x/T$

$$LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x}\alpha^{x}}{(1-\hat{\pi})^{T-x}\hat{\pi}^{x}} \sim \chi^2_1$$

- Christoffersen: $LR_{ind} \sim \chi^2_1$ compares $\hat{\pi}_{01}$, the breach rate after a quiet day, with $\hat{\pi}_{11}$, the rate after a breach
- Joint test: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2_2$ (full formulas: Step 4 of the case study)
- S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS (historical simulation) 1.67%, FHS (filtered historical simulation) 1.23%, GARCH-EVT (GARCH filter with a tail from extreme value theory, EVT) 1.15% breaches
- On the S&P 500 and the BET, a GARCH filter plus a flexible tail (FHS, GARCH-EVT) wins

Chapter 9: Realised Volatility

- Intraday prices measure daily variance (RV, realised variance)
- The log-HAR (heterogeneous autoregressive) model captures its long memory Corsi (2009)

$$\ln RV_{t+1} = b_0 + b_d \ln RV_t + b_w \overline{\ln RV}_t^{(5)} + b_m \overline{\ln RV}_t^{(22)} + u_{t+1}$$

- $\overline{\ln RV}_t^{(5)}$, $\overline{\ln RV}_t^{(22)}$: the mean of $\ln RV$ over the last 5 and 22 days (week and month)
- SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by $\sqrt{RV_t}$
- log-HAR beats GARCH(1,1)-t in QLIKE by 10% for SPY ($p = 0.021$) and 22% for Bitcoin

Chapter 10: Market Microstructure

- Prices form in limit order books; the spread pays for processing, inventory and adverse selection
- Amihud ratio (basis points, bp, per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

Chapter 11: Continuous-Time Models

- Brownian motion is the limit of random walks; Itô's lemma solves the price equations
- S&P 500 against geometric Brownian motion (GBM): 62 days beyond ± 4 standard deviations, against 0.59 expected
- Hill index of S&P 500 losses below 4 (Chapter 16): the fourth moment is likely infinite, unlike the Normal log returns of GBM

Chapter 12: Options and the Volatility Surface

- Black–Scholes prices options by hedging; implied volatility is the market's view of σ Black & Scholes (1973)
- 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
- The variance risk premium is positive on 86% of days, but not in crashes

Chapter 13: Machine Learning in Finance

- Flexible models, but little signal and much noise; the finance-specific part is the process López de Prado (2018)
- Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
- S&P 500: the fractionally differenced series is stationary at $d^* = 0.25$; no model beats "always up"

Chapter 14: Deep Learning and Time-Series Foundation Models

- LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
- SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH-t 0.283; the first two are not significantly different
- Chronos-2 VaR 1% without recalibration on the S&P 500: 1.67% breaches

Chapter 15: LLMs and Sentiment Analysis

- From dictionaries to FinBERT and large language models (LLMs): text becomes a number
- Qwen 7B vs 1.5B on financial headlines: $+21.5$ percentage points in accuracy, McNemar $p < 0.001$
- A headline-sentiment strategy: Sharpe $+0.63$ before and -1.48 after costs; break-even cost 1.5 bp per trade

Chapter 16: Digital Assets and DeFi

- The chapter case study (Pele et al. (2023)): cryptocurrencies form a distinct asset class in a feature space where a tail factor separates them
- DeFi (decentralised finance) runs on smart contracts
- Bitcoin–S&P 500 link: sup-Wald 20.2 (the largest Wald break statistic over candidate dates; 5% critical value 11.6), break estimated on 9 Mar 2020, not at the 2024 ETF launch
- Hill tail index of losses: Bitcoin 2.73, 95% CI $[2.12, 3.33]$; S&P 500 2.87, 95% CI $[2.07, 3.67]$
- The intervals overlap: Bitcoin tails are not significantly heavier than those of the S&P 500

Chapter 17: Bubbles and Crashes

- A bubble is a price path that grows faster than fundamentals can justify
- Bitcoin 2014–2026 (weekly): GSADF (Generalised Sup Augmented Dickey–Fuller) (the largest right-tailed unit-root statistic over all windows) 3.85 against a 95% critical value of 2.27: explosive episodes are detected
- One vs two Markov regimes for Bitcoin: likelihood-ratio statistic $LR = 123.2$ against a parametric-bootstrap 95% critical value of 11.8 ($p = 0.005$)

Chapter 18: Systemic Risk and Networks

- Systemic risk: losses of one institution that spread to the whole system
- ΔCoVaR 1% (the change in the system's VaR 1% when a bank moves from its median to distress) of eleven banks: 4 of 25 within-region pairs differ at 5%, 0 after Holm (a correction for testing many pairs at once: pitfall of multiple testing)
- Overlap of intervals alone would have ranked none
- Simulated LRMES (Long-Run Marginal Expected Shortfall) (22 days, market below -10%): JPMorgan 10.6% today, 23.7% with the information of 31 March 2020

Pitfalls of Design: Look-Ahead Bias and Overfitting (1/2)

- **Look-ahead bias**: any parameter, threshold or scaling estimated on the full sample leaks the future
  - Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set; the case study measures the effect on BET
  - Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
  - Remedy: rolling or expanding windows; purging and embargo; write the information set $\mathcal{F}_{t-1}$ next to every forecast

Pitfalls of Design: Look-Ahead Bias and Overfitting (2/2)

- **Overfitting**: the in-sample fit rewards noise that will not repeat
  - Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
  - Backtest overfitting: trying many strategies and keeping the best Bailey et al. (2017)
  - The deflated Sharpe ratio corrects for the number of trials and for non-normal returns Bailey & López de Prado (2014)
  - Remedy: fix the design before looking at the test period
  - Report every model you tried
  - Compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall of Design: Multiple Testing (1/2)

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$, and the best reaches $|t| \approx 3.06$
  - This is why the factor literature asks for $|t| > 3$ Harvey, Liu & Zhu (2016)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery

Pitfall of Design: Multiple Testing (2/2)

- Remedy: count your tests and correct
  - With m tests, ordered p-values $p_{(1)} \leq \cdots \leq p_{(m)}$ and level a (e.g. 5%), reject $H_{(i)}$ if

Bonferroni:	$p_{(i)} \leq a/m$
Holm:	$p_{(j)} \leq a/(m-j+1)$ for every $j \leq i$
Benjamini–Hochberg:	$i \leq \max\{k : p_{(k)} \leq k\, a/m\}$

  - Bonferroni and Holm Holm (1979) control the chance of any false discovery (FWER)
  - Holm rejects everything Bonferroni rejects, sometimes more, at the same FWER
  - Benjamini–Hochberg (BH) Benjamini & Hochberg (1995) controls the expected share of false discoveries among the rejections (FDR)
  - For forecasts: the model confidence set (MCS) instead of many pairwise tests Hansen, Lunde & Nason (2011)

Pitfall of Data: Aligning Returns Instead of Prices

- Bitcoin trades every day, SPY only on weekdays
  - Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
  - Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- Bitcoin, 2020–2026, on the common days: cumulative return 1 025% (right) against 680% (wrong)
  - Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong alignment hides weekend risk
- Remedy: align prices on common days first; one series alone stays on its own calendar

Pitfall of Inference: Heavy Tails Ignored (1/2)

- Classical standard errors assume independence and finite fourth moments
  - BET losses: Hill tail index $\hat{\alpha} = 2.39$, block-bootstrap 95% CI $[2.12, 3.03]$ Hill (1975)
  - Moments of order $\geq \alpha$ are infinite: with $\alpha < 4$ the population kurtosis is infinite
  - So the sample excess kurtosis (11.1) estimates nothing, and a bootstrap interval for it is not valid Athreya (1987): report the tail index instead
- BET GARCH persistence $p = \alpha + \beta$ (here α, β: the GARCH(1,1) parameters): the Wald CI [0.974, 1.007] crosses $p = 1$, the boundary of the parameter region $p \leq 1$ that we impose
  - If the true p may be 1, the Normal approximation fails Andrews (1999)
  - Profile-likelihood CI (all p whose likelihood-ratio statistic, maximising over the other parameters, is below the critical value) inside the parameter space $p \leq 1$: $[0.975, 1]$ (IGARCH, $p = 1$, not rejected against the boundary critical value 2.71, the 5% point of a 50:50 mixture of 0 and χ^2_1), so the half-life is at least 27 days and has no finite upper bound

Pitfall of Inference: Heavy Tails Ignored (2/2)

- Remedy: HAC (heteroskedasticity and autocorrelation consistent) errors, block bootstrap where it is valid, tail indices instead of kurtosis, boundary-aware intervals
- The Newey–West HAC standard error of a mean $\bar{d}$ Newey & West (1987): $\mathrm{SE}(\bar{d}) = \sqrt{\hat{\Omega}/T}$, with

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{L}\Big(1 - \frac{j}{L+1}\Big)\hat{\gamma}_j$$

  - $\hat{\gamma}_j$: the sample autocovariance at lag j; L: the number of lags included
  - The Bartlett weights $1 - j/(L+1)$ guarantee $\hat{\Omega} \geq 0$
  - This is the long-run variance in the Diebold–Mariano test (Step 5)

Persistence Near 1: Which Boundary? (1/2)

- $p \leq 1$ is a restriction we choose, not a property of GARCH: its interior, with persistence strictly below one, gives a finite unconditional variance and a finite half-life
  - Equality is the IGARCH boundary
  - Without it, p slightly above 1 is admissible: the process can still be strictly stationary, when $E \ln(\alpha z_t^2 + \beta) < 0$ Nelson (1990), but it has no finite variance and no half-life

Persistence Near 1: Which Boundary? (2/2)

- The BET estimate $\hat{p} = 0.9907$ is inside the region; the difficulty is the **true** p, which may equal 1
  - At $p = 1$ the estimator is not approximately Normal, so the Wald interval [0.974, 1.007] is uninformative under our restriction, rather than "wrong"
- Under the restriction, the likelihood-ratio test of $H_0 : p = 1$ has the boundary distribution $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ Andrews (1999)
  - 5% critical value 2.71 instead of 3.84; BET: $LR = 1.01$ with a p-value of 0.16, so IGARCH ($p = 1$) is not rejected
  - Boundary critical values apply only when the true parameter can sit on the edge of the space you imposed, not whenever an estimate is close to some bound

Pitfall of Evaluation: a Loss That Cannot Rank the Target (1/2)

- Variance forecasts are compared with a noisy proxy (r_t^2 or RV_t)
  - For a variance forecast f_t and a proxy $\hat{\sigma}_t^2$: the MSE (mean squared error) loss $(\hat{\sigma}_t^2 - f_t)^2$ and QLIKE $\hat{\sigma}_t^2/f_t - \ln(\hat{\sigma}_t^2/f_t) - 1$
  - With a conditionally unbiased proxy, MSE and QLIKE rank forecasts as the true variance would
  - QLIKE depends only on the ratio proxy/forecast, so volatile days weigh less than under MSE Patton (2011)
  - Chapter 9: log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive

Pitfall of Evaluation: a Loss That Cannot Rank the Target (2/2)

- ES alone has no strictly consistent scoring function on the usual broad classes of distributions
  - The pair (VaR, ES) has one, under mild conditions Fissler & Ziegel (2016)
  - Use the FZ0 score (Step 5) with Diebold–Mariano tests Diebold & Mariano (1995)
- Remedy: choose the loss before the comparison, and state it in the project report

One Question, Four Chapters (1/2)

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6 670 daily log returns
- Log return in %: $r_t = 100 \ln(P_t / P_{t-1})$, loss $L_t = -r_t$
- Duplicate records (rows identical to the previous one) and weekend-dated rows removed
  - Trading days with an unchanged close kept

One Question, Four Chapters (2/2)

The five steps

- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student-t innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones
- Formal comparison of the four models: FZ0 loss, Diebold–Mariano (DM) tests, model confidence set

One Symbol, Four Meanings: α in the Case Study

Where	Meaning of α	Value	Enters
Step 1, Hill	tail index of losses	2.39	$P(L > x)$ decays like $x^{-\alpha}$
Step 2, GARCH	weight of yesterday's squared shock	0.192	$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
Steps 3–4, VaR and Kupiec	VaR level: promised breach probability	1%	LR_{uc}, DQ
Step 5, FZ0 and ES	ES level used to score the pair (VaR, ES)	2.5%	$S(v, e; y)$

- Before substituting a number, ask which α the formula means: the Kupiec test needs $\alpha = 0.01$, never 2.39 or 0.192
- The letter p also has two meanings here: the GARCH persistence $p = \alpha + \beta$ and a p-value

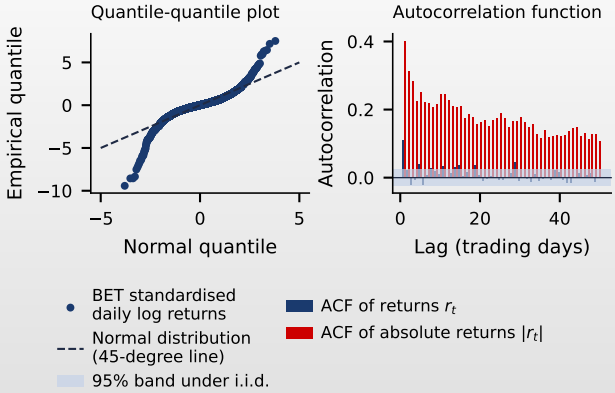
Step 1: What We Measure and How We Test It

- Tails: the Hill tail index $\hat{\alpha}$ of losses (formula: the Chapter 1 review), with a block-bootstrap 95% CI
- Volatility clustering: the autocorrelations of $|r_t|$ and the Ljung–Box test on r_t^2

$$Q(m) = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k} \;\sim\; \chi^2_m \text{ under } H_0$$

  - H_0: the first m autocorrelations are zero; $\hat{\rho}_k$: the sample autocorrelation at lag k
  - On r_t^2 the test detects volatility clustering; here $m = 10$, so the 5% critical value is 18.3
- Next chart: left, quantiles of the standardised returns against those of the Normal distribution; right, the ACF of r_t and of $|r_t|$

Step 1: Stylised Facts of the BET (1/3)



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Step 1: Stylised Facts of the BET (2/3)

- Heavy tails (left, quantiles against the Normal distribution)
  - Hill tail index of losses $\hat{\alpha} = 2.39$, 95% CI [2.12, 3.03] (block bootstrap, $k = 153$ largest losses)
  - 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution
  - Worst daily loss: -13.1%, on 7 Jan 2009

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Step 1: Stylised Facts of the BET (3/3)

- Volatility clustering (right)
  - ACF (autocorrelation function) of $|r_t|$ at lag 1: 0.40, slow decay; the band assumes i.i.d. (independent, identically distributed) returns
  - Ljung–Box $Q(10)$ on r_t^2: 2 440 (5% critical value 18.3)
- A Normal, constant-volatility VaR: too low in stress, too high in calm periods

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Step 1: Reading the Stylised Facts

- The kurtosis is not reported as an estimate, because $\hat{\alpha} < 4$ (pitfall of inference)
- Autocorrelation of the returns themselves: 0.11
  - Typical of a less liquid market (Chapters 2, 10)

Step 2: the GARCH(1,1)-t Model

- Today's variance is a weighted sum of a constant, yesterday's squared shock and yesterday's variance Bollerslev (1986)

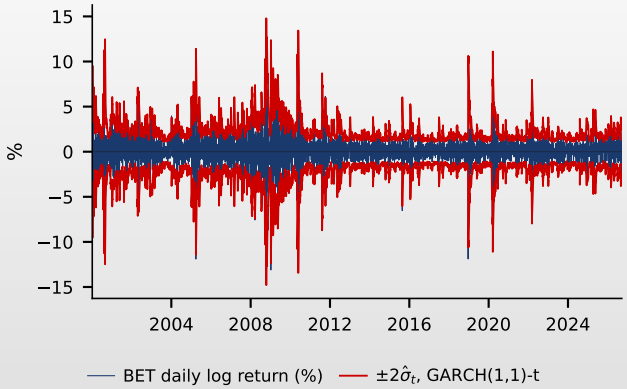
$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$$

  - r_t: daily log return; μ: mean; ε_t: shock; σ_t: conditional volatility, known at $t-1$; z_t: i.i.d. Student-t with ν degrees of freedom, scaled to variance 1
  - $\omega > 0$, $\alpha, \beta \geq 0$: α measures the reaction to news, β the memory; ν small: heavy tails ($\nu \to \infty$ gives the Normal distribution)
- Persistence $\alpha + \beta < 1$: covariance stationary; a variance shock decays by the factor $\alpha + \beta$ each day

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$$

  - $h_{1/2}$: half-life, the number of days after which half of a variance shock is gone; $\alpha + \beta = 1$ (IGARCH): shocks never fade
- Estimated by maximum likelihood, robust (sandwich) standard errors

Step 2: GARCH(1,1)-t Volatility



- BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student-t innovations, 2000–2026
  - Robust standard errors: $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$: heavy tails even after filtering
- Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$, a half-life of about 74 trading days; it cannot be bounded away from 1 (pitfall of inference)
- Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample $\hat{\sigma}_T = 1.67\%$ (18 Sep 2026), and the one-step forecast for the next day is $\hat{\sigma}_{T+1} = 1.76\%$

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Step 2: Today's Volatility and the Long-Run Level (1/2)

- Two numbers at the end of the sample: $\hat{\sigma}_T$ is known at $T-1$; $\hat{\sigma}_{T+1}$ also uses the last return $r_T = 2.17\%$ and is the input of tomorrow's VaR

$$\hat{\sigma}^2_{T+1} = \hat{\omega} + \hat{\alpha}\,(r_T - \hat{\mu})^2 + \hat{\beta}\,\hat{\sigma}^2_T = 0.0448 + 0.192\,(2.17 - 0.08)^2 + 0.799 \times 1.67^2 = 3.11$$

  - So $\hat{\sigma}_{T+1} = \sqrt{3.11} = 1.76\%$: a large last return lifts the forecast above $\hat{\sigma}_T = 1.67\%$

- Long-run variance implied by the model

$$\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta} = \frac{0.0448}{0.0093} = 4.82$$

  - Returns are in %, so $\bar{\sigma}^2$ is in $\%^2$: $\bar{\sigma} = 2.20\%$ a day, 34.7% a year using the observed 249.82 returns per year, against a sample standard deviation of 1.40% a day (variance 1.96) and 22.1% a year

Step 2: Today's Volatility and the Long-Run Level (2/2)

- Why the gap is not a contradiction: the denominator $1 - \alpha - \beta = 0.0093$ is small and imprecise
  - Holding ω fixed, $p = 0.9867$ gives $\bar{\sigma}^2 = 3.37$ and $p = 0.9947$ gives 8.47; the profile CI reaches $p = 1$, where $\bar{\sigma}^2$ is infinite
  - Use $\hat{\sigma}_{T+1}$ and the short-horizon forecast path; treat the long-run level as weakly identified
- The forecast path returns geometrically to $\bar{\sigma}^2$, at the rate $\hat{\alpha} + \hat{\beta}$

$$\hat{\sigma}^2_{T+h} = \bar{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^{h-1}\left(\hat{\sigma}^2_{T+1} - \bar{\sigma}^2\right), \qquad h = 21\text{:}\ \ 4.82 + 0.9907^{20}\,(3.11 - 4.82) = 3.40$$

  - So $\hat{\sigma}_{T+21} = 1.84\%$: after one month the forecast has covered only 17% of the distance to the long-run level

Step 3: Four VaR 1% Forecasts (1/2)

- Unconditional, on the last 500 days: one quantile for the whole window
  - **HS** (historical simulation): the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
  - **Normal**: $\mathrm{VaR} = -(\hat{\mu} + \hat{\sigma} z_{\mathbf{0.01}})$; $\hat{\mu}$, $\hat{\sigma}$: sample mean and standard deviation; $z_{\mathbf{0.01}} = -2.326$: the 1% quantile of the standard Normal distribution
- Conditional, GARCH-t on the last 1000 days: the quantile scales with today's volatility forecast $\hat{\sigma}_t$

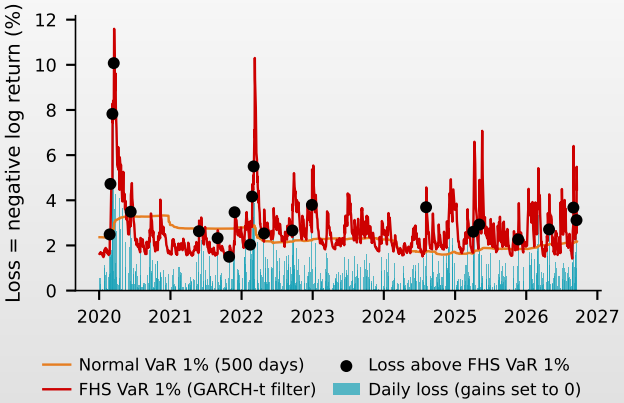
$$\textbf{GARCH-}t\text{:}\ \ \mathrm{VaR}_t = -\left(\hat{\mu} + \hat{\sigma}_t\, t_\nu^{-\mathbf{1}}(0.01)\sqrt{(\nu-2)/\nu}\right), \qquad \textbf{FHS}\text{:}\ \ \mathrm{VaR}_t = -\left(\hat{\mu} + \hat{\sigma}_t\, \hat{q}_{\mathbf{0.01}}(z)\right)$$

  - $t_\nu^{-\mathbf{1}}(0.01)$: the 1% quantile of a Student-t with ν degrees of freedom; $\sqrt{(\nu-2)/\nu}$ rescales it to variance 1
  - **FHS** (filtered historical simulation): $\hat{q}_{\mathbf{0.01}}(z)$, the empirical 1% quantile of the standardised residuals $z_t = (r_t - \hat{\mu})/\hat{\sigma}_t$, replaces the Student-$t$ quantile McNeil & Frey (2000)

Step 3: Four VaR 1% Forecasts (2/2)

- Real-time design
  - Forecast for day t uses data up to $t-1$ only
  - GARCH re-estimated every 21 trading days; $\hat{\sigma}_t$ updated daily
- Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
  - A breach (hit) on day t: the loss $-r_t$ exceeds VaR_t; expected breaches at 1%: $0.01\,T \approx 54$
- Unconditional models react only as the window rolls; conditional ones react through $\hat{\sigma}_t$, within days

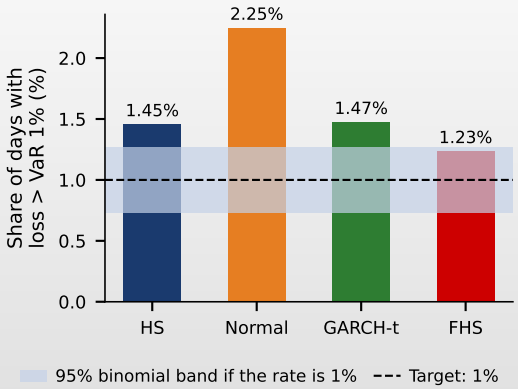
Step 3: Losses and VaR 1%, 2020–2026



- Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR
- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
  - The Normal VaR on a 500-day window moves only slowly
- 1 675 days, about 17 breaches expected: FHS 22, Normal 41
  - Normal breaches cluster in the first weeks of every stress episode
- Last VaR forecast: FHS 4.48%, Normal 2.18%

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Step 4: Breach Rates



- Share of days with a loss above the VaR 1% forecast, over the $T = 5\,432$ evaluation days of Step 3
- Light-blue band: the 95% binomial range [0.74%, 1.27%] if the true rate is 1%
- Only FHS (1.23%) lies inside the band; the Normal model (2.25%) breaches more than twice as often as it should
- Next
  - are the differences significant?
  - are the breaches independent?

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Step 4: the Three Backtests

- Kupiec coverage test Kupiec (1995): is the breach rate equal to $\alpha = 1\%$?

$$LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x}\alpha^{x}}{(1-\hat{\pi})^{T-x}\hat{\pi}^{x}} \sim \chi^2_1, \qquad \hat{\pi} = x/T$$

  - x: number of breaches in T days; $\hat{\pi}$: observed breach rate; $LR_{uc} = 0$ when $\hat{\pi} = \alpha$; reject at 5% when $LR_{uc} > 3.84$

- Christoffersen independence test Christoffersen (1998): is a breach today more likely after a breach yesterday?

$$LR_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}}\,\hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\,\hat{\pi}_{01}^{n_{01}}\,(1-\hat{\pi}_{11})^{n_{10}}\,\hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1$$

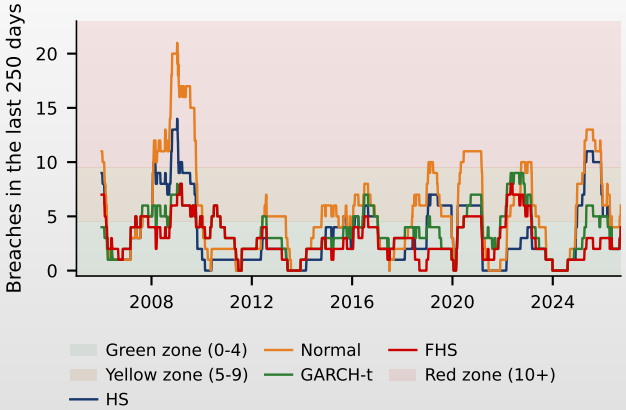
  - n_{ij}: number of days in state j after a day in state i (1 = breach, 0 = no breach)
  - $\hat{\pi}_{01} = n_{01}/(n_{00}+n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10}+n_{11})$: breach rates after a quiet day and after a breach
  - In LR_{ind}, $\hat{\pi} = (n_{01}+n_{11})/(n_{00}+n_{01}+n_{10}+n_{11})$, the breach rate over the $T-1$ transitions; it differs from the Kupiec x/T by at most $1/(T-1)$
  - Joint test: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2_2$ (5% critical value 5.99); the p-values p_{uc}, p_{ind}, p_{cc} come from these χ^2 distributions

Step 4: Coverage and Independence Tests

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

- Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- Only FHS is rejected by none of the three tests at 5%; HS and Normal also fail independence: their breaches cluster
- GARCH-t: independence is not rejected, but there are too many breaches: the Student-t tail is not heavy enough at 1%

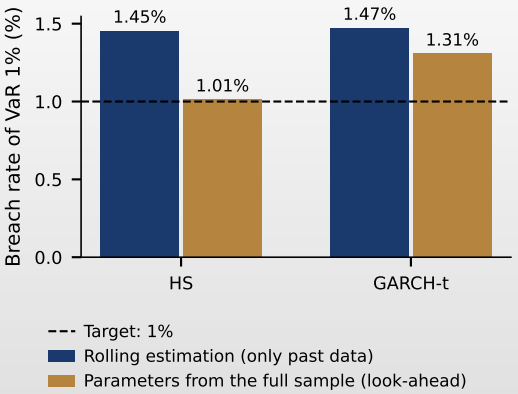
Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- Normal and HS enter the red zone in episodes, when their breaches cluster (shares: last column of the Step 4 table)
- The two GARCH-filtered models never enter it

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The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction
- GARCH-t with full-sample parameters: 1.31% of days, with rolling estimation 1.47%
- Both look better than their real-time versions: the gap is the value of information nobody had
- Only the rolling result is evidence about a forecasting model

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Step 5: Three Tools for a Formal Comparison (1/2)

- **Dynamic Quantile (DQ)** test Engle & Manganelli (2004): are the breaches predictable?
  - Regress $I_t - 0.01$ on a constant, the four lagged hits $I_{t-1}, \ldots, I_{t-4}$ and VaR_t; $I_t = 1$ if the loss exceeds VaR_t, else 0
  - DQ: Wald statistic that all six coefficients are zero, $DQ \sim \chi^2_6$ under a correct model
  - It detects breaches predictable from the past or from the level of VaR, which the Christoffersen test (one lag) can miss
- **FZ0 score** Patton, Ziegel & Chen (2019): one number per day that scores a (VaR, ES) forecast pair; lower is better

$$S(v, e; y) = -\frac{1\{y \le v\}(v - y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$$

  - $\alpha = 2.5\%$: the ES level; $v = -\mathrm{VaR}$, $e = -\mathrm{ES}$: the forecasts written as return quantiles ($e \le v < 0$); y: the realised return; $1\{\cdot\}$: 1 if the condition holds, else 0
  - Strictly consistent for (VaR, ES) Fissler & Ziegel (2016): on average the true pair has the lowest score

Step 5: Three Tools for a Formal Comparison (2/2)

- Comparing the FZ0 scores of two models
  - Pairwise Diebold–Mariano tests Diebold & Mariano (1995): $t = \bar{d}/\sqrt{\hat{\Omega}/T}$, d_t the FZ0 difference of the two models on day t, $\hat{\Omega}$ its HAC (Newey–West) long-run variance
  - $t \approx N(0,1)$ under equal accuracy
  - Model confidence set at 10% Hansen, Lunde & Nason (2011): the set of models that contains the best one with probability 90%
  - p_{MCS}: a model stays in the set while $p_{MCS} \geq 0.10$
- **ES backtest**
  - Comparative, against HS as the standard model; red if the model is significantly worse, green if significantly better, yellow otherwise Nolde & Ziegel (2017)

Step 5: Which Forecast Pair Does FZ0 Score?

- A second forecast pair from the same four models, at the Basel ES level $\alpha = 2.5\%$: VaR 2.5% and ES 2.5%
  - Not the VaR 1% of Steps 3–4: FZ0 is consistent only when v and e are the quantile and the tail mean at the same level α
- Each model's ES 2.5% is the mean loss beyond its own VaR 2.5%, from the same loss distribution it uses for VaR in Step 3
  - **HS**: the average of the window losses beyond the empirical 2.5% quantile
  - **Normal**: $\mathrm{ES} = -\hat{\mu} + \hat{\sigma}\,\varphi(z_{\mathbf{0.025}})/0.025 = -\hat{\mu} + 2.338\,\hat{\sigma}$, with φ the standard Normal density
  - **GARCH-t**: $-\hat{\mu}$ plus $\hat{\sigma}_t$ times the upper-tail mean of the Student-t rescaled to variance 1
  - **FHS**: the negative of the sum of the conditional mean and $\hat{\sigma}_t$ times the mean of standardised residuals z_t at or below their 2.5% quantile

Step 5: The ES 2.5% Forecasts in Numbers

Model	Average VaR 1% (Steps 3–4)	Average ES 2.5% (Step 5)
HS	3.66%	3.94%
Normal	2.90%	2.92%
GARCH-t	2.95%	3.08%
FHS	3.17%	3.41%

- Averages of the daily forecasts over the $T = 5\,432$ evaluation days, losses in % of the position
- Check that the levels differ: for the Normal model, ES 2.5% / VaR 1% $= 2.338/2.326 \approx 1.005$
  - The table gives $2.92/2.90 = 1.005$: ES 2.5% is close to VaR 1%, while an ES 1% would be about 15% higher
- The gap between the columns grows with the weight of the tail the model allows: about 7.5% for HS and FHS (empirical tails), 4% for GARCH-t, 0.5% for the Normal

Step 5: Results

Model	DQ	p_{DQ}	FZ0	p_{MCS}	ES zone
HS	192.1	<0.001	1.424	0.00	standard
Normal	364.6	<0.001	1.495	0.00	red
GARCH-t	17.9	0.007	1.175	0.18	green
FHS	9.9	0.129	1.157	1.00	green

- Only FHS passes the DQ test at 5% ($p = 0.129$); GARCH-t fails it ($p = 0.007$) through its excess of breaches
- FZ0 (lower is better): FHS and GARCH-t are not significantly different (DM $t = 1.39$), both beat HS and Normal ($|t| > 4.9$)
  - The 10% model confidence set is {GARCH-t, FHS}
- Comparative ES backtest: Normal red, GARCH-t and FHS green
- The GARCH-filtered models are superior; FHS remains the only model rejected by no test (coverage, independence, DQ)

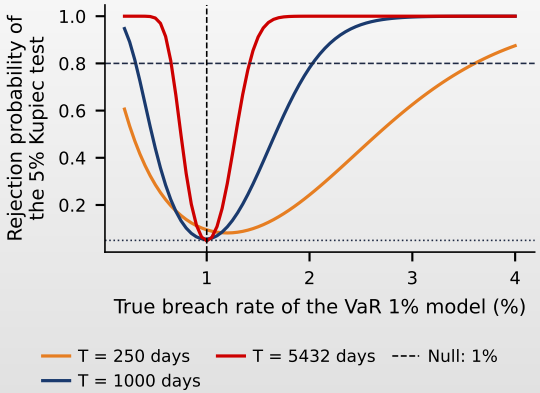
The Power of a Backtest: Definition

- The question: if the VaR 1% model is wrong, how often does the Kupiec test notice?
- Power at the true breach rate π: the exact binomial probability of the rejection region R_T of the 5% test

$$\text{Power}(\pi) = \sum_{x \in R_T} \binom{T}{x} \pi^x (1-\pi)^{T-x}, \qquad R_T = \{x : LR_{uc}(x) > 3.84\}$$

  - At $\pi = 1\%$ the same sum gives the actual size of the test
  - At $T = 250$, R_T holds 0 breaches and any number from 7 up
- Minimum detectable effect (MDE): the smallest true rate $\pi > 1\%$ with $\text{Power}(\pi) \geq 80\%$
- Power grows with T: the standard error of the observed rate $\hat{\pi}$ is $\sqrt{\pi(1-\pi)/T}$

How Much Can a Backtest Detect? The Power of Kupiec



- Exact binomial probability that the 5% Kupiec test rejects a VaR 1% model, by its true breach rate
- $T = 250$ (one Basel year): a true rate of 2% is detected only 24% of the time
- MDE, searched on a grid with steps of 0.05 percentage points: 3.65% at $T = 250$, 2.05% at $T = 1\,000$, 1.45% at $T = 5\,432$
  - Even at $T = 5\,432$, a true 1.3% is detected with probability 55% only
- A non-rejection with low power is "no evidence", not "no problem": state the power before the test

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The Project in Four Steps (1/2)

- An empirical study of one question about financial markets, answered with the methods of this course
  - **70%** of the final grade (quizzes and seminar activity 20%, attendance 10%, no final exam)
- **1. Replicate**: reproduce one full table or figure of a published paper (or of the lecture), cell by cell
  - An AI assistant cannot guess a published table (standard: next slide)

The Project in Four Steps (2/2)

- **2. Extend**
  - New data, a new market (the BVB, crypto) or a more recent sample, under a **pre-analysis plan** recorded in a dated commit before the test period is opened
  - The plan states the null, the test, the loss, the sample split, the identification strategy, the power of the main test and the specification grid with its correction
- **3. Deliver on GitHub**: a repository that regenerates every number (standard: next slide)
  - `AI_USE.md` and `AI_ERRORS.md` are mandatory
- **4. Present and defend**: team presentation, then a 10-minute oral defence
  - Each member explains one result and answers a "what changes if..." question; grades can differ between members

Replication Standard

- Target: one full table or figure, with its DOI, page and table number, not a single coefficient
  - Report a cell-by-cell comparison: published value, your value, difference, and the reason for each gap (sample, filter, data vendor, definition)
- Large-scale replications show why: many anomalies shrink out of sample or with better data Hou, Xue & Zhang (2020), while most factors replicate once the design is right Jensen, Kelly & Pedersen (2023)
- Reproducibility: a lockfile of package versions, fixed random seeds, series names and dates of the data, one command that regenerates every number
  - Different teams on the same data disagree more than their standard errors suggest Menkveld et al. (2024): more reproducible work is associated with a smaller gap (research case study, this chapter)

Choosing the Question (1/2)

A good question is

- **Specific**: one market, one period, one quantity
- **Testable**: a null hypothesis and a benchmark
- **Feasible**
  - Data you can obtain and clean in a few weeks (the course data sets, in the course GitHub repository; public sources: the National Bank of Romania, BNR, the Federal Reserve Economic Data, FRED, Kenneth French Data Library, DefiLlama)
- **Relevant**: an answer matters to an investor, a bank or a regulator

Choosing the Question (2/2)

Examples

- Too vague: "Is Bitcoin risky?"
  - Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin?"
  - Then: "Which model forecasts it best since 2024?"
- Too broad: "Can machine learning predict the market?"
  - Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability	Chapters 1–2: moments, robust tests, variance ratios
Expected returns, factors, portfolios	Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR
Risk measures and capital	Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with machine learning (ML), deep learning or text	Chapters 13–15: purged cross-validation (CV), foundation models, sentiment
Crypto, bubbles, systemic risk	Chapters 16–18: stablecoins, GSADF, CoVaR, networks

- Starting points: the Part C question of every seminar (table in the Appendix)
  - Your own question is welcome if it passes the five questions of this chapter

Identification: from Correlation to a Claim (1/2)

- Event questions ("did spot ETFs, the FTSE upgrade or a bank tax change X?", FTSE: the index provider FTSE Russell) need a counterfactual
  - A control market not hit by the event: difference-in-differences on the risk measure Y (e.g. volatility or VaR 1%)

$$\hat{\delta} = \left(\bar{Y}^{\text{after}}_{\text{treated}} - \bar{Y}^{\text{before}}_{\text{treated}}\right) - \left(\bar{Y}^{\text{after}}_{\text{control}} - \bar{Y}^{\text{before}}_{\text{control}}\right)$$

  - Equivalently: δ in $Y_{it} = a + b\,D_i + c\,P_t + \delta\,D_i P_t + \varepsilon_{it}$, with $D_i = 1$ for the treated market, $P_t = 1$ after the event and HAC standard errors
  - Identifying assumption: parallel trends, i.e. without the event Y would have moved alike in both markets
  - Its check: before the event, Y moves in parallel in the two markets
  - A synthetic control built from several untreated markets Abadie, Diamond & Hainmueller (2010)
  - Placebo event dates (e.g. $\pm$ one year)

Identification: from Correlation to a Claim (2/2)

- Unknown timing: a sup-Wald break test Andrews (1993); report the estimated break date, not an imposed one
- Forecasting questions: out of sample only, with a test period declared in the plan
- One event is $n = 1$: say what the design can and cannot identify

Power and Robustness without p-Hacking (1/2)

- **Power**: before the main test, compute the smallest effect the design detects with 80% power at 5% (MDE)
  - For an approximately Normal estimate with standard error SE, a two-sided 5% test and 80% power

$$\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\mathrm{SE} = 2.80\,\mathrm{SE}, \qquad T_{\min} = \left(\frac{2.80\,\sigma}{\delta}\right)^2 \text{ when } \mathrm{SE} = \sigma/\sqrt{T}$$

  - z_p: the p-quantile of the standard Normal distribution; δ: the effect you look for; σ: the standard deviation of one observation
  - Deliverable: a simulated power curve of the main test at your T
  - Short crypto, ETF or post-event samples make many questions underpowered (Kupiec at $T = 250$: MDE 3.65%)

Power and Robustness without p-Hacking (2/2)

- Specification curve: estimate every defensible combination of window, lags, α and sample start
  - Plot all estimates sorted, with their intervals Simonsohn, Simmons & Nelson (2020)
  - Report the spread across specifications, not the best cell
- Correct the family: Holm for any false discovery, Benjamini–Hochberg for their share, or White's bootstrap reality check for data snooping White (2000)
  - Significance thresholds must rise with the number of tests tried Harvey & Liu (2020), Harvey (2017)

Forecast Evaluation for Risk Projects

- Volatility: a robust loss (MSE or QLIKE) against a conditionally unbiased proxy (pitfall of evaluation)
  - Many models: the model confidence set Hansen, Lunde & Nason (2011), not the lowest average loss
- VaR: coverage, independence and the DQ test Engle & Manganelli (2004); with a benchmark, conditional predictive ability tests Giacomini & White (2006)
  - State the power of every backtest at your T
- ES: score the pair (VaR, ES) with FZ0; comparative backtests with three zones Nolde & Ziegel (2017); dynamic (VaR, ES) models Patton, Ziegel & Chen (2019)
- The integrated case study (Steps 1–5) is the template

The Report and the Presentation (1/2)

The report: an 8–12-page paper

- **1. Introduction**: question, hypothesis, contribution
- **2. Literature**: 10–15 clickable references, at least 3 from the last five years, including Franke, Härdle & Hafner (2019), McNeil, Frey & Embrechts (2015)
- **3. Data and replication**: sources, period, cleaning rules; the replicated table cell by cell
- **4. Design**: plan, identification, power, benchmark, loss, tests
- **5. Results** with their uncertainty
- **6. Robustness**: specification curve with the correction
- **7. Conclusion**: the answer in one sentence, and its limits

The Report and the Presentation (2/2)

The presentation (about 12 minutes): the slides, in order

- Question (1 slide)
- Data and one chart that shows the problem (2)
- Method and benchmark (2)
- Main result with its uncertainty (2–3)
- Robustness, limits, answer (2)
- One message per slide; readable axes, legend below the chart
- Every member presents and can answer questions on the code; then the oral defence, without AI

Project Grading Criteria

Component	Grading criteria
Replication (step 1)	A published table or figure reproduced cell by cell; every gap explained
Question, data, extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Plan, identification, power, real-time design, benchmark, specification curve with a correction
Interpretation and report	Economic meaning, limitations, honest conclusion; 8–12 pages, 10–15 references
GitHub repository (step 3)	One command reproduces every result (lockfile, seeds); `AI_USE.md`, `AI_ERRORS.md`
Presentation and defence (step 4)	Individual: one result explained, answers to "what changes if..."

Deliverables and Timeline

- Week 1 after this chapter: team, question, one-page proposal (Seminar 19, C2), GitHub repository
- Weeks 2–3: data, replication (step 1), extension plan recorded in a dated commit
- Weeks 4–5: extension, evaluation, robustness (step 2)
- Final weeks: the 8–12-page report, repository with README and the two AI files (step 3), presentation and oral defence (step 4)
- Exact dates are announced at the course and seminar

The AI Policy and the Two Mandatory Files

Policy

- **Allowed**: AI assistants for code, debugging, literature search and writing
- **Declared**: undeclared use of AI counts as plagiarism
- **Checked**: you are responsible for every line; a reference that does not exist counts as fabricated data
- **No AI during the oral defence**: "the AI wrote it" is not an explanation

The two files

- `AI_USE.md`: tools and versions, who used them for what, where AI-written code or text was kept, how you verified the replicated table, data, formulas and references
- `AI_ERRORS.md`: at least **three** places where the assistant was wrong: what you asked, what it answered, what was wrong, how you found out, the correct version
- Errors you caught are a strength of the project, not a weakness

Typical AI Errors in Financial Modelling (1/2)

- **Look-ahead**: a rolling statistic or a scaler that includes day t in the forecast for day t
  - Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- **Sign of VaR**: the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
  - Check: $\mathrm{VaR}_\alpha = -q_\alpha(r)$, with $q_\alpha(r)$ the α-quantile of returns, is positive whenever $q_\alpha(r) < 0$; a one-day VaR 1% of an index is a few percent
- **Annualisation**: volatility multiplied by 252 instead of $\sqrt{252}$, or 252 days used for crypto that trades every day
  - Check: over h periods, means scale with h, volatilities with $\sqrt{h}$ (i.i.d. returns); use the real number of trading days (365 for crypto)

Typical AI Errors in Financial Modelling (2/2)

- **Random cross-validation (CV) on time series**
  - Shuffled K-fold trains on the future (pitfall of design, Chapter 13)
  - Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo López de Prado (2018)
- **Invented references**, wrong degrees of freedom, library defaults you did not choose, numbers stated without running code
  - Check: search every DOI on Crossref and find the claim in the paper, as in the AI-critique exercise of every seminar (Part C)

How Much Do Researchers' Choices Matter? (1/2)

- Menkveld et al. (2024), *Journal of Finance* 79(3): 164 research teams test the same six hypotheses on the same data (Finance Crowd Analysis Project, #fincap)
- **Standard error**
  - Uncertainty from sampling, the data-generating process
  - **Nonstandard error**: uncertainty from the analysis path each researcher picks, the evidence-generating process
  - Model: team j reports $X_j = e_j + \varepsilon_j$, e_j the researcher fixed effect of its path, ε_j the sampling error
  - Nonstandard error $:= Q_{0.75}(x) - Q_{0.25}(x)$, the interquartile range (IQR) of the estimates across teams, $Q_p(x)$ being the p-quantile of the team estimates x
  - The IQR, not the standard deviation, because the e_j can have fat tails

How Much Do Researchers' Choices Matter? (2/2)

- **Data**
  - All trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records
  - Six hypotheses of "no annual trend" (market efficiency, realised spread, client share of volume, ...)
  - Each team reports the average yearly change in % with its standard error (SE)
  - Four stages in 2021: independent analysis, peer feedback, the five best papers shared, final estimates; design details in the ▸ Appendix
- Link to this chapter: every team project is one path through the "garden of forking paths"
  - Pre-registration, the specification grid and a repository that reruns every number are the remedies

Key Results: Size and Quality (1/2)

- **Size**
  - For market efficiency the median estimate is -1.1% per year, with an IQR of 6.7 percentage points (pp)
  - Even for the client share of volume, a simple ratio, the IQR is 1.2 pp around -3.3% (Table I, Panel C)
  - The median SE of 2.5% implies a Gaussian IQR of $1.35 \times 2.5 = 3.4\%$ (the IQR of a Normal distribution is 1.35 standard deviations), half the 6.7 pp
  - The no-dispersion null is rejected for all six hypotheses, $p < 0.0001$ (Table II)

Key Results: Size and Quality (2/2)

- **Quality**
  - A rise of one standard deviation in reproducibility is associated with a 25.0% lower IQR, the same rise in peer rating with a 33.3% lower IQR (Table III, Fig. 2)
  - Code that reruns end to end is associated with answers closer to the consensus

Key Results: Feedback, Beliefs and Forks (1/2)

- **Feedback**: from stage 1 to stage 4 the IQR falls by 47.2% and the interdecile range by 68.2%
  - Peer review narrows the dispersion but does not remove it
- **Beliefs**: pooled over hypotheses, the teams' predicted standard deviation of estimates is 71.7% below the realised one
  - Researchers underestimate how much teams analysing the same data can differ

Key Results: Feedback, Beliefs and Forks (2/2)

- **Forks**
  - Model choice and sampling frequency move the estimates most
  - A relative-change model gives a median annual trend of $-23\,000\%$ on daily data, -200% monthly and -4.56% yearly
  - Log differences add up exactly across frequencies and a linear trend is linear in the data, so neither has this frequency wedge
  - Jensen's inequality ($\mathbb{E} \ln M_t \leq \ln \mathbb{E} M_t$, with $M_t = X_t / X_{t-1}$) biases the mean relative change of a positive measure upwards; the negative values in the paper need another mechanism

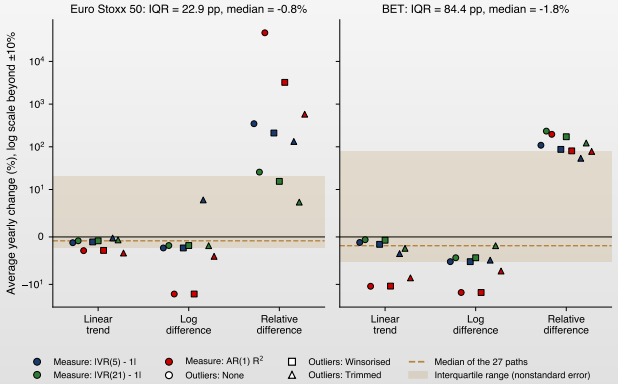
A Mini-Multiverse on Course Data: the Design

- Daily closes 2002–2018 from the course data
  - Forks: outliers × measure × model, 27 paths
  - The band is the IQR, the nonstandard error of the multiverse
- Measures of inefficiency
  - $|VR(q) - 1|$ ($VR(q)$: the variance ratio of Chapter 2, $q = 5, 21$ days) and the R^2 of an autoregressive AR(1) model: departures from a random walk
  - A positive trend means declining efficiency

MFM_ch19_nonstandard_errors

A Mini-Multiverse on Course Data: 27 Paths per Index

Legend: $VR(q)$, the q-day variance ratio; AR(1), the first-order autoregressive model; IQR, the interquartile range; pp, percentage points.



MFM_ch19_nonstandard_errors

A Mini-Multiverse on Course Data: Results (1/2)

- IQR: 22.9 pp (Euro Stoxx 50) and 84.4 pp (BET)
  - Around medians of -0.8% and -1.8% a year

A Mini-Multiverse on Course Data: Results (2/2)

- The model fork creates the dispersion
  - Over the nine linear-trend paths the IQR is only 2.0 pp for the Euro Stoxx 50 and 7.5 pp for the BET
  - Relative differences explode: medians $+210.4\%$ and $+108.7\%$, up to $46\,823\%$
  - The paper also reports explosions of relative changes, but negative ones
  - Our measures are non-negative, so here they explode upwards
- Sampling benchmark: the Gaussian IQR, $1.35\times$ the median SE (25.9 and 50.8 pp)
  - The IQR exceeds this benchmark for the BET, not for the Euro Stoxx 50

Why Relative Changes Explode: a Toy Example (1/2)

- A measure that returns to its start: $m = (1,\ 0.01,\ 1)$
  - Relative changes $m_t/m_{t-1} - 1$: -99%, then $+9\,900\%$; their mean is $+4\,900.5\%$
  - Log changes: $\ln 0.01$, then $\ln 100$; their mean is exactly 0, the right answer for a measure with no trend
  - For a positive measure Jensen's inequality pushes the mean relative change up (Appendix): the course multiverse explodes upwards

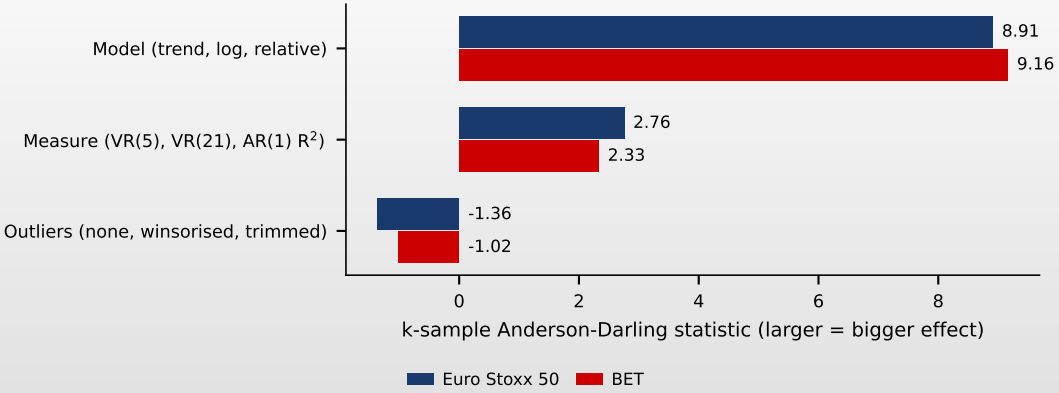
Why Relative Changes Explode: a Toy Example (2/2)

- A positive measure can never fall by more than 100%
  - A yearly trend such as $-23\,000\%$ therefore needs another mechanism: a measure that changes sign near zero ($m_{t-1} = 0.001$, $m_t = -0.2$ gives $-0.2/0.001 - 1 = -20\,100\%$), or daily changes summed rather than compounded into a yearly rate (250 daily changes of -1% add up to -250%)
- Lesson for your plan: inspect denominators near zero and justify the transformation
  - Pre-specify log differences for positive measures or a linear trend

Which Fork Moves the Estimates? The Statistic

- k-sample Anderson–Darling statistic of the 27 estimates, grouped by the option at each fork
  - It compares the distributions of the k groups; larger means more separated
  - Standardised, so it is centred at 0 when the groups are equal: a negative value (outliers below) means less separation than chance
- A descriptive ranking, not a test
  - The 27 estimates share the same data

Which Fork Moves the Estimates?



MFM_ch19_nonstandard_errors

Which Fork Moves the Estimates? Results (1/2)

- Euro Stoxx 50 and BET
  - Model 8.91 and 9.16; measure 2.76 and 2.33
  - Outliers -1.36 and -1.02
- Same ranking as the paper: model choice moves the estimates most

Which Fork Moves the Estimates? Results (2/2)

- Holds on a large euro-area index and on the BET, a frontier-market index in 2002–2018 (FTSE Russell: Secondary Emerging since September 2020)
- For your project
  - Fix the model of the yearly change in the plan (log differences or linear trends)
  - Report the whole grid and the fork that drives it

Limits and an Open Extension (1/2)

- **Limits stated by the authors**
  - The multiverse covers a few forks only and explains the reported estimates poorly
  - Beliefs were asked as standard deviations, not IQRs
  - The test of dispersion assumes consistent reported SEs; the dynamics of journal refereeing are kept out of scope

Limits and an Open Extension (2/2)

- **Limits of our replication**
  - A multiverse spans only the options in its grid; its IQR neither estimates nor bounds from below the dispersion across real teams
  - Daily closes only, so the paper's intraday frequency pairs cannot be run
- **Open extension**: does a pre-registered grid shrink the nonstandard error of a risk question?
  - Measure the IQR of VaR 1% conclusions for the BET across defensible paths

An Open Question

- **Open question**: how much of a result is the data, and how much the researcher?
  - Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries Jensen, Kelly & Pedersen (2023)
  - AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
  - Formal version: the conclusion must hold over a pre-registered set $\mathcal{S}$ of defensible specifications, with the family corrected (FWER or FDR)
  - Report its distribution over $\mathcal{S}$, not the best cell

The Discovery Loop and Its Checks (1/2)

- **Loop**
  - Question and H_0/H_1 (week 1)
  - Literature and the replication target seen in the PDF (weeks 1–2)
  - Pre-registration of the extension, recorded in a dated commit before the first extension result (week 2)
  - Replication as a passing test in the repository (weeks 2–3)
  - The whole grid (weeks 4–5)
  - An AI examiner before the defence

The Discovery Loop and Its Checks (2/2)

- **Checks by the researcher**
  - The target number is in the paper
  - The plan precedes the first results file in the commit history
  - Every run is logged and counted
  - A fresh clone regenerates every number
  - Each member can derive the key formula without AI
  - Prompts for every step and the test for each check: Appendix

Mini-Case: Replicate, Then Extend (1/2)

- **Replicate**: FHS VaR 1% for the BET; Chapter 8 reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
  - The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$)
  - Same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal, reading each p-value alone at 5%: "only GARCH-EVT is correctly calibrated on both markets"

Mini-Case: Replicate, Then Extend (2/2)

- **Extend**
  - The 12 conditional-coverage tests of Chapter 8 (S&P 500 and BET $\times$ 6 models) form one family
  - Bitcoin and EUR/RON stay outside the family: on these markets some GARCH forecasts are missing (Chapter 8)
  - Rejections at 5%: 8 without correction, 6 with Holm (FWER), 7 with Benjamini–Hochberg, BH (FDR 5%)
  - FHS fails only on the S&P 500 ($p = 0.040$); within the family of 12 this rejection is no longer significant, so after Holm FHS and GARCH-EVT pass on both markets
- Interpretation: "FHS fails for the S&P 500" depends on whether the test is read alone or in its family
- In the project, fix the family in the plan before seeing results and report both readings

Your Turn: the Capstone Plan

- **Question**: your project idea from Part C, written as H_0/H_1, with the observation that would falsify it
- **Replicate first**: one published table seen in the paper, turned into a passing test in the repository
  - Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- **Deliverables** along the timeline: proposal and replication target, pre-registration commit, passing replication test, grid of results, mock defence, report, oral defence
- **AI log**: the prompts that changed the design in `AI_USE.md`; every caught error, with the test that caught it, in `AI_ERRORS.md` (templates are in the project folder)

Key Takeaways

- Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- Heavy tails change inference: use robust errors and the bootstrap, report intervals, state the power
- Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- The analysis path is itself a source of uncertainty: pre-register, replicate, report the whole grid
- New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Check Your Understanding (1/2)

Questions

- Why does the Kupiec test need $\alpha = 0.01$, not 2.39 or 0.192?
- Why is the sample excess kurtosis of BET returns not an estimate of anything?
- Why is the Wald interval for GARCH persistence uninformative once we restrict to $p \leq 1$, and what replaces it?
- What does a non-rejection of the Kupiec test at $T = 250$ tell you?
- Why is the IQR of a 27-path multiverse not the nonstandard error of 164 teams?

Check Your Understanding (2/2)

Further study

- Seminar 19: the exercises A1–A12, B1–B8 and C1–C3 apply this chapter to the BET
- Textbooks: Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)
- Advanced courses on quantinar.com: Statistics of Financial Markets
  - Measuring Statistical Risk
- Code and data: Quantlets/Ch_19 (GitHub)

Two Computations on New Numbers

Try it

- **1.** A VaR 1% model has $x = 6$ breaches in $T = 400$ days. Compute LR_{uc} with the Kupiec formula of Step 4 and decide at 5%
- **2.** A GARCH(1,1) has $\alpha + \beta = 0.97$. Compute the half-life $h_{1/2}$; how does it change if $\alpha + \beta = 0.985$?

Check your answer

- **1.** $\hat{\pi} = 6/400 = 1.5\%$

$$LR_{uc} = 2\left[(394 \ln 0.985 + 6 \ln 0.015) - (394 \ln 0.99 + 6 \ln 0.01)\right] = 0.88$$

  - Below 3.84: not rejected; at $T = 400$ the test has little power, so this is "no evidence", not "no problem"
- **2.** $\ln 0.5 / \ln 0.97 = 22.8$ days and $\ln 0.5 / \ln 0.985 = 45.9$ days
  - Halving $1 - \alpha - \beta$ roughly doubles the half-life

Chapter 0: Financial Markets in 2026

- Data quality is a modelling decision: source, adjustment and calendar change the result (Chapter 0)
- Mean and volatility estimated from i.i.d. log returns, annualised with the number of periods a year, with their standard errors (se)

$$\mathrm{se}(\hat{\mu}_{\mathbf{ann}}) = \frac{\sigma_{\mathbf{ann}}}{\sqrt{Y}}, \qquad \mathrm{se}(\hat{\sigma}) \approx \sigma\sqrt{\frac{K-1}{4T}}$$

  - Y: sample length in years, whatever the frequency; T: number of daily returns
  - σ, $\sigma_{\mathbf{ann}}$: daily and annualised volatility
  - K: population kurtosis (a finite fourth moment is needed)
  - Serial dependence calls for long-run variances
- SPY 2015–2026: $\mathrm{se}(\hat{\mu}_{\mathbf{ann}}) \approx 5.1\%$ for a CAGR of 13.7%, i.e. an annualised mean log return $\hat{\mu}_{\mathbf{ann}} = 100\ln(1.137) = 12.8\%$: means are barely estimable, volatilities are

Back

Chapter 1: Stylised Facts and Returns (1/2)

- Heavy tails and volatility clustering in every market Cont (2001) (Chapter 1)
- Ljung–Box test of no autocorrelation up to lag m

$$Q(m) = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k} \sim \chi_m^2$$

  - $\hat{\rho}_k$: sample autocorrelation at lag k
  - T: sample size
  - The χ_m^2 reference needs i.i.d. returns
  - Under volatility clustering it over-rejects
  - Use the robust Q^* with summands $T\hat{\rho}_k^2/\hat{\tau}_k$, $\hat{\tau}_k$ the estimated variance factor of $\sqrt{T}\hat{\rho}_k$ (about 1 for i.i.d. returns) Lobato, Nankervis & Savin (2001)

Back

Chapter 1: Stylised Facts and Returns (2/2)

- Tails: Hill estimator $\hat{\alpha}$ of the tail index, $\text{se} \approx \hat{\alpha}/\sqrt{k}$ Hill (1975)
  - k: number of largest losses used
  - If $\alpha < 4$ the sample kurtosis is not consistent and its bootstrap fails Athreya (1987)

Back

Chapter 2: Market Efficiency

- Efficiency varies over time (adaptive markets); tests need robust inference (Chapter 2)
- Heteroskedasticity-robust variance ratio test Lo & MacKinlay (1988)

$$VR(q) = \frac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}, \qquad z^*(q) = \frac{\widehat{VR}(q) - 1}{\widehat{\mathrm{se}}^*} \Rightarrow N(0,1)$$

  - q: horizon in periods; $VR = 1$ under a random walk, $VR > 1$: positive autocorrelation, $VR < 1$: mean reversion
  - $\widehat{\mathrm{se}}^*$: heteroskedasticity-robust standard error; $z^* \Rightarrow N(0,1)$ under a martingale difference null
- Several horizons q at once: the joint test $\max_q |z^*(q)|$ with studentised maximum modulus critical values Chow & Denning (1993)
  - Rolling windows need a multiplicity correction too

Back

Chapter 3: Factor Models and Asset Pricing (1/2)

- Hundreds of published factors require higher hurdles Harvey, Liu & Zhu (2016) (Chapter 3)
- GRS test that all pricing errors (intercepts) are zero Gibbons, Ross & Shanken (1989)

$$\mathrm{GRS} = \frac{T-N-K}{N} \frac{\hat{\alpha}^\top \hat{\mathbf{\Sigma}}^{-1} \hat{\alpha}}{1+\hat{\mu}_f^\top \hat{\mathbf{\Omega}}^{-1} \hat{\mu}_f} \sim F(N, T-N-K)$$

  - T periods, N assets, K factors; $\hat{\alpha}$: the N intercepts (alphas); $\hat{\mu}_f$: factor means; $\hat{\mathbf{\Sigma}}$, $\hat{\mathbf{\Omega}}$: residual and factor covariances, divisor T
  - Exact F only under i.i.d. Normal errors; a large GRS means the factors do not price the assets

Back

Chapter 3: Factor Models and Asset Pricing (2/2)

- Fama–MacBeth s.e. ignore that betas are estimated: Shanken's covariance for estimated factor loadings Shanken (1992)

$$\hat{V}_{\mathbf{Sh}} = (1+c)(\hat{V}_{FM} - \hat{\Sigma}_f/T) + \hat{\Sigma}_f/T, \qquad c = \hat{\gamma}^\top \hat{\Sigma}_f^{-1} \hat{\gamma}$$

  - $\hat{\gamma}$: estimated factor risk premia
  - $\hat{\Sigma}_f$: factor covariance
  - $\hat{V}_{FM}$: Fama–MacBeth covariance of $\hat{\gamma}$
  - $c \geq 0$ inflates it, more when premia are large relative to factor risk

Back

Chapter 4: Portfolio Optimisation

- Shrinkage, constraints and risk-based rules stabilise the weights; 1/N is a hard benchmark DeMiguel, Garlappi & Uppal (2009) (Chapter 4)
- Standard error of the Sharpe ratio $SR = (\mu - r_f)/\sigma$ per period, i.i.d. Normal returns Lo (2002)

$$\operatorname{se}(\widehat{SR}) \approx \sqrt{(1 + \widehat{SR}^2/2)/T}$$

  - μ, σ: mean and volatility per period; r_f: risk-free rate; T: number of periods; serial correlation changes the annualisation factor
- Is portfolio A better than 1/N? Test $SR_A - SR_B = 0$ with HAC or studentised bootstrap inference Ledoit & Wolf (2008), not by comparing two point estimates

Back

Chapter 5: Conditional Volatility: GARCH

- Persistence close to 1; asymmetry in equities; heavy-tailed innovations (Chapter 5)
- QMLE (quasi-maximum likelihood) of GARCH: the sandwich covariance Bollerslev & Wooldridge (1992)

$$\sqrt{T}(\hat{\theta} - \theta) \Rightarrow N\left(0, A^{-1} B A^{-1}\right)$$

  - $\theta = (\omega, \alpha, \beta)$: parameters of $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; A: expected Hessian of the Normal log-likelihood (sign changed); B: variance of its score
  - $A = B$ when the innovations are Normal; Hessian-only s.e. (A^{-1}) are wrong for non-Normal innovations
- If we impose $\alpha + \beta \leq 1$ and the true persistence may equal 1, Wald intervals are uninformative
  - Use profile likelihood or bootstrap restricted to the parameter space Andrews (1999)

Back

Chapter 6: Multivariate Volatility and Dependence

- Copulas separate the margins from dependence; tail dependence drives joint losses (Chapter 6)
- DCC is estimated in two steps: second-step s.e. must account for the first-step volatilities Engle (2002)
- Crisis vs calm correlation: ρ rises mechanically with volatility; test on the adjusted correlation Forbes & Rigobon (2002)

$$\rho^* = \frac{\rho}{\sqrt{1 + \delta(1 - \rho^2)}}$$

  - ρ: correlation measured in the crisis; δ: relative rise in the variance of the source market from calm to crisis; $\rho^* < \rho$ when $\delta > 0$

Back

Chapter 7: Value-at-Risk and Expected Shortfall

- Filtered historical simulation and conditional EVT react to volatility in days McNeil & Frey (2000) (Chapter 7)
- Historical VaR, i.i.d. returns with a continuous density $f(q_\alpha) > 0$

$$\sqrt{T}(\hat{q}_\alpha - q_\alpha) \Rightarrow N\big(0,\ \alpha(1-\alpha)/f(q_\alpha)^2\big)$$

  - q_α: the α-quantile of returns, $\mathrm{VaR}_\alpha = -q_\alpha$; $\hat{q}_\alpha$: the empirical quantile; f: return density; in the tail f is small, so the estimate is noisy
  - Under dependence $\alpha(1-\alpha)$ becomes the long-run variance of the hits $1\{r_t \le q_\alpha\}$ (1 if the condition holds, 0 otherwise)
- ES averages the αT largest losses: with αT of a few dozen and a tail index near 3, its variance is dominated by one or two days

Back

Chapter 8: Backtesting and Evaluating Risk Forecasts

- Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests Fissler & Ziegel (2016) (Chapter 8)
- VaR backtests: the number and the timing of breaches (formulas: Step 4 of the case study)
  - Kupiec $LR_{uc} \sim \chi^2_1$ (is the breach rate α?); Christoffersen $LR_{ind} \sim \chi^2_1$ (is a breach independent of yesterday's?)
  - DQ test: regress $I_t - \alpha$ on lagged hits and VaR_t, $I_t = 1$ if the loss exceeds VaR_t; $DQ \sim \chi^2_k$, k the number of regressors Engle & Manganelli (2004)
- Power is low at $T = 250$ (the power of Kupiec, case study); compare models with conditional predictive ability tests Giacomini & White (2006)

Back

Chapter 9: Realised Volatility

- Returns divided by realised volatility have a distribution close to the Normal distribution
- HAR errors are heteroskedastic and, for multi-day targets that overlap, serially correlated: use HAC s.e.
  - Forecasts are compared with DM tests on a robust loss Patton (2011)
- Microstructure noise biases *RV* (the sum of squared intraday returns) upwards at high frequency: two-scale estimators remove the bias Zhang, Mykland & Aït-Sahalia (2005)

Back

Chapter 10: Market Microstructure

- Liquidity follows the clock and the VIX, and disappears in stress
- The Roll spread: the bid–ask bounce makes consecutive price changes negatively correlated

$$s_{\text{Roll}} = 2\sqrt{-\text{Cov}(\Delta p_t, \Delta p_{t-1})}$$

  - Δp_t: change of the transaction price; when the sample covariance is positive the estimator is undefined, so report how often
- An illiquidity premium is a cross-sectional claim: Fama–MacBeth with Newey–West s.e. and the Shanken correction Shanken (1992)

Back

Chapter 11: Continuous-Time Models

- Jumps explain the tails, stochastic volatility the clustering (Chapter 11)
- Normality of GBM log returns: Jarque–Bera $JB = \frac{T}{6}\left(S^2 + (K-3)^2/4\right) \sim \chi^2_2$ rejects everywhere
  - S: sample skewness; K: sample kurtosis; both 0 and 3 under the Normal distribution
  - Jumps: $(RV - BV)$ standardised $\Rightarrow N(0,1)$ Barndorff-Nielsen & Shephard (2006)
  - BV: bipower variation, a sum of products of adjacent absolute intraday returns, robust to jumps
- The test is daily: over many days, control the number of false jump days

Chapter 12: Options and the Volatility Surface

- Smile and skew encode the fear of crashes (Chapter 12)
- The variance risk premium

$$VRP_t = VIX_t^2 - RV_{t,t+21}$$

  - VIX_t^2: the risk-neutral expected variance over the next month; $RV_{t,t+21}$: realised variance over the next 21 trading days
  - Consecutive values overlap, so the mean needs Newey–West s.e. with at least 21 lags
- Implied volatility is a risk-neutral forecast: the premium, not a forecast error, explains most of the gap

Back

Chapter 13: Machine Learning in Finance

- Purged cross-validation, meta-labelling and deflated Sharpe ratios Bailey & López de Prado (2014) (Chapter 13)
- Selection among N strategies: the deflated Sharpe ratio tests $SR > E[\max_N \widehat{SR}]$ under the null of no skill Bailey & López de Prado (2014)
  - $E[\max_N \widehat{SR}]$: the expected best estimated Sharpe ratio among N strategies whose true Sharpe ratio is zero; it grows with N
- Cross-validation on time series must purge overlapping labels and embargo the test fold, or its accuracy is biased upwards

Back

Chapter 14: Deep Learning and Time-Series Foundation Models

- Evaluate them with Chapter 8 tools, after the release of their weights (Chapter 14)
- DM test on QLIKE differences with HAC variance Diebold & Mariano (1995)

$$t_{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}} \Rightarrow N(0,1)$$

  - d_t: QLIKE of model 1 minus QLIKE of model 2 on day t, $\bar{d}$ its mean over T days
  - $\hat{\Omega}$: HAC long-run variance of d_t
  - Valid for non-nested comparisons
- With many candidate models, the model confidence set (MCS) controls the family-wise error Hansen, Lunde & Nason (2011)

Back

Chapter 15: LLMs and Sentiment Analysis

- Text embeddings enter forecasting models as features
- Predictive regression $r_{t+1} = a + bS_t + \varepsilon_{t+1}$
  - S_t: sentiment score of day t
  - b: predictive slope
  - A persistent S_t whose innovations correlate with returns biases $\hat{b}$ and the t-test Stambaugh (1999)
- Classifier accuracies on the same texts are paired: compare them with the exact McNemar test, not two separate intervals

Back

Chapter 16: Digital Assets and DeFi

- Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance
- Crypto factor models: GRS and Fama–MacBeth with the Shanken correction Gibbons, Ross & Shanken (1989), Shanken (1992)
- The standard errors of these tests need HAC or bootstrap versions, because of heavy tails
- Tails: Hill indices with intervals, not sample kurtosis (Chapter 1)
- A break in a correlation is tested with an unknown date: the sup-Wald statistic is the largest Wald statistic over candidate break dates Andrews (1993)

Back

Chapter 17: Bubbles and Crashes

- Right-tailed unit-root tests date the start and the end of explosive episodes (Chapter 17)
- Bubble tests: the largest right-tailed ADF statistic over many windows

$$GSADF = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2}$$

  - $ADF_{r_1}^{r_2}$: the ADF (augmented Dickey–Fuller) statistic on the sample fraction $[r_1, r_2]$; r_0: minimum window; SADF (Sup Augmented Dickey–Fuller) fixes $r_1 = 0$
  - Under the unit-root null, SADF and GSADF converge to functionals of Brownian motion; critical values depend only on r_0 Phillips, Shi & Yu (2015)
- Volatility shifts distort the size: use wild-bootstrap critical values; control the multiplicity of date-stamping

Back

Chapter 18: Systemic Risk and Networks (1/2)

- CoVaR (the VaR of the system conditional on the distress of one institution), SRISK (the capital shortfall of a bank in a systemic crisis) and spillover networks measure the contribution of each institution (Chapter 18)
- ΔCoVaR: the change in the system's VaR when institution i moves from its median to distress

$$\Delta\text{CoVaR}^i = \text{CoVaR}_\alpha^{sys|X^i=-\text{VaR}_\alpha^i} - \text{CoVaR}_\alpha^{sys|X^i=\textbf{median}}$$

  - X^i: return of institution i
  - $\text{CoVaR}_\alpha^{sys|\cdot}$: VaR α of the system return given the state of i
  - Differences between banks: paired tests on joint block-bootstrap draws with a Holm correction
  - At $\alpha = 1\%$ only about 40 tail days, so extremal-quantile issues

Back

Chapter 18: Systemic Risk and Networks (2/2)

- SRISK: LRMES (the expected loss of a bank when the market falls by more than 10% over 22 days) simulated from GJR–GARCH (the asymmetric Glosten–Jagannathan–Runkle GARCH) + DCC Brownlees & Engle (2017)
  - Connectedness tables need bootstrap intervals and shrinkage vector autoregressions (VARs)

Back

Project Ideas: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR (BET total return) autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	Does an optimised portfolio of BVB blue chips, in RON, beat 1/N out of sample?
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	How much capital does a BVB portfolio need compared with an S&P 500 portfolio?

- Each seminar gives the data, a starting method and a reference analysis discussed in class

Back

Project Ideas: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?
15	Does headline sentiment pay after costs?
16	Is Bitcoin an alternative asset, a risk asset or a hedge?
17	Is there an AI bubble?
18	How exposed are Romanian banks to euro-area banking stress?

- Chapter 20 (realised volatility with signatures, VOLARE data) also suits a forecasting project

Back

Turning an Idea into a Project

- Replicate first: one published table or figure from a paper behind the idea (step 1)
- Narrow it: one market, one period, one number to forecast or test
  - Example: Seminar 8 becomes "Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 portfolio of the BET and EUR (EUR/RON exchange rate) in 2015–2026?"
- Extend it: a second market, a new model, a longer or more recent sample
  - Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

Back

Data Conventions for the Project

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
  - Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
  - Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
  - Remove holiday rows where the vendor repeated the previous price, and weekend quotes where the market is closed; annualise with the real frequency
  - Several assets: align prices on common trading days first, then compute returns
- Look for erroneous prices (bad ticks) before modelling: jumps of $\pm 50\%$ in adjusted prices are usually data errors

Back

Honest Evaluation Checklist (1/2)

Before you look at the results

- ⊡ Write the hypothesis, the benchmark and the loss function
- ⊡ Fix the estimation window and the test period
- ⊡ Decide how many models and parameters you will try, and the correction for them
- ⊡ Write the identification strategy and the power of the main test

Back

Honest Evaluation Checklist (2/2)

When you report

- ⊡ Out-of-sample results only for claims about forecasting
- ⊡ Every headline number with a standard error, CI or p-value
- ⊡ Costs for trading strategies; subperiods and a second market for robustness
- ⊡ What did **not** work, and why

Back

Where Projects Usually Lose Points

- A question that changes after the results are known
- In-sample fit presented as forecasting power
- No benchmark, or a weak one chosen to be beaten
- Numbers without uncertainty; many tests without correction
- Charts without units, legends inside the data, unexplained acronyms
- Code that does not run, or numbers in the report that the code does not produce
- AI output kept without checks: undeclared use, unchecked code, references that do not exist

Back

Nonstandard Errors, Data and Method (1/3)

- **Data**
  - All trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records, each side flagged as client or own account (Sec. I.A)
  - Six hypotheses, each "no annual trend" in: market efficiency, realised bid–ask spread, client share of volume, realised spread on client orders, share of market orders among client orders, gross trading revenue of clients
  - Each team chooses its own measure of each quantity

Back

Nonstandard Errors, Data and Method (2/3)

- **Four stages** in 2021 (Sec. I.A)
  - Independent analysis (paper and code)
  - Feedback from two peer evaluators
  - The five best papers shared
  - Final estimates without the code constraint
  - A pre-analysis plan was filed publicly before the experiment started (Sec. I.A)

Back

Nonstandard Errors, Data and Method (3/3)

- **Quality**
  - Team quality (first principal component of five proxies), reproducibility score 0–100 by the certification agency Cascad, demeaned peer rating (Sec. I.B)
  - Quantile regressions of the estimates at the 10th, 25th, 50th, 75th and 90th percentiles; a shrinking IQR means less dispersion (Sec. I.C)
- **Inference**
  - Each estimate is tested against the median estimate, with a Bonferroni correction at a family level of 0.5% (Sec. I.C)
  - Multiverse analysis: the teams report their choice at each fork of Table V
  - The k-sample Anderson–Darling statistic ranks the forks by how much they move the estimates (Sec. II.C)

Back

Nonstandard Errors, Replication Details

- **Forks from Table V**
  - Outliers (none; the annual measure winsorised or trimmed at its 2.5 and 97.5 percentiles), measure (variance ratio day to week, variance ratio day to month, R^2 of an autoregressive AR(1) model), model (linear trend, log difference, relative difference)
  - Daily data allow the day-to-week and day-to-month comparisons ($VR(5)$, $VR(21)$); the intraday measure frequencies of Table V need intraday data
- **Why relative differences explode**: $m_t = |VR - 1|$ or R^2 is close to zero in some years (paper: Fig. 7)
  - No trend, stationary m_t: the linear slope and the mean log change are centred at zero, but for independent m_t, $\mathbb{E}(m_t / m_{t-1}) = \mathbb{E}m \cdot \mathbb{E}(1/m) \geq 1$ (Jensen), so the mean relative change is positive
- **Anderson–Darling statistic**: the 27 estimates are strongly correlated, so tabulated critical values and p-values do not apply

Back

The Discovery Loop, Prompts (1/4)

- **Week 1, the question**: sharpen a project idea into H_0/H_1 with a replication target
  - `"Here is our project idea from Part C of Seminar N. Give three sharper, testable versions, the published table each would replicate and the series from the course data it needs."`
- **Weeks 1–2, literature and target**: the number to replicate is one you have seen in the PDF
  - `"List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample."`
  - Resolve every DOI on Crossref; search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves

Back

The Discovery Loop, Prompts (2/4)

- **Week 2, hypotheses and pre-registration**, recorded in a dated commit before the first result
  - "For our H0 and H1, list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction."
  - The commit history of the repository is the evidence that the plan came first

Back

The Discovery Loop, Prompts (3/4)

- **Weeks 2–3, code and replication**: the replicated table becomes a test in the repository
  - `"Write a function that reproduces every cell of the published table from the course data and an assertion with the tolerance we accept."`
- **Weeks 4–5, extension and robustness**: run the whole pre-registered grid
  - `"Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?"`

Back

The Discovery Loop, Prompts (4/4)

- **Before the defence, critique**: the assistant asks, the team answers
  - "`Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them.`"
  - The team answers without AI, as at the defence itself
- **Report**: the distribution of results over the grid, not the best cell; prompts in `AI_USE.md`, caught errors in `AI_ERRORS.md`

Back

What the Researcher Must Check (1/2)

- **The replication target exists**: number, table and sample taken from the paper, not from the assistant's summary
  - Test: the report cites the page and the table; a second member finds the same number there
- **The plan came first**
  - Test: in the commit history, the pre-registration of the extension precedes the first extension results file

Back

What the Researcher Must Check (2/2)

- **Forking paths**: more specifications tried than reported
  - Test: every run writes its configuration and result to a log; the report states their number and corrects the family
- **Reproducibility**: numbers entered manually in the report
  - Test: a fresh clone and one command regenerate every number to the reported digit
- **Understanding**: each member can derive the key formula and defend every choice
  - Test: an internal mock defence, without AI, one week before the real one

Back

Acronyms used in this material (1/3)

- **ACF**: AutoCorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ASE**: Academia de Studii Economice din București — *the Bucharest University of Economic Studies*
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAGR**: Compound Annual Growth Rate
- **CAPM**: Capital Asset Pricing Model
- **CI**: Confidence Interval
- **CoVaR**: Conditional Value-at-Risk (of the system, given an institution)
- **COVID-19**: COronaVIrus Disease 2019
- **CV**: Cross-Validation
- **DCC**: Dynamic Conditional Correlation
- **DeFi**: Decentralised Finance
- **DM**: Diebold–Mariano (test)
- **DOI**: Digital Object Identifier
- **DQ**: Dynamic Quantile (test)

Acronyms used in this material (2/3)

- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **EVT**: Extreme Value Theory
- **FDR**: False Discovery Rate
- **FHS**: Filtered Historical Simulation
- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-EVT**: GARCH filter combined with Extreme Value Theory for the tail
- **GBM**: Geometric Brownian Motion
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GRS**: Gibbons–Ross–Shanken (test)
- **GSADF**: Generalised Sup Augmented Dickey–Fuller (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HS**: Historical Simulation
- **IGARCH**: Integrated GARCH
- **IQR**: InterQuartile Range
- **JB**: Jarque–Bera (test)

Acronyms used in this material (3/3)

- **LR**: Likelihood Ratio
- **LRMES**: Long-Run Marginal Expected Shortfall
- **MCS**: Model Confidence Set
- **MDE**: Minimum Detectable Effect
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **PDF**: Portable Document Format
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **RV**: Realised Variance
- **SADF**: Sup Augmented Dickey–Fuller (test)
- **SE**: Standard Error
- **SML**: Security Market Line
- **SPY**: SPDR S&P 500 ETF Trust
- **SR**: Sharpe Ratio
- **SRISK**: Systemic RISK (capital shortfall measure)
- **TLT**: iShares 20+ Year Treasury Bond ETF
- **US**: United States
- **VIX**: CBOE Volatility IndeX
- **VOLARE**: VOLatility Archive for Realized Estimates (database)
- **VR**: Variance Ratio
- **VRP**: Variance Risk Premium

References (1/6)

- Abadie, A., Diamond, A., Hainmueller, J. (2010). Synthetic control methods for comparative case studies: estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490), 493–505.
- Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- Andrews, D.W.K. (1999). Estimation when a parameter is on a boundary. *Econometrica*, 67(6), 1341–1383.
- Athreya, K.B. (1987). Bootstrap of the mean in the infinite variance case. *Annals of Statistics*, 15(2), 724–731.
- Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Barndorff-Nielsen, O.E., Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300.
- Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

References (2/6)

- Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.
- Brownlees, C., Engle, R.F. (2017). SRISK: a conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1), 48–79.
- Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.
- Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Engle, R. (2002). Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

References (3/6)

- Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Fissler, T., Ziegel, J.F. (2016). Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). Applied Quantitative Finance, 3rd ed. Springer.
- Harvey, C.R. (2017). Presidential address: the scientific outlook in financial economics. *Journal of Finance*, 72(4), 1399–1440.

References (4/6)

- Harvey, C.R., Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5), 2503–2553.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lo, A.W., MacKinlay, A.C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.

References (5/6)

- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- McNeil, A.J., Frey, R., Embrechts, P. (2015). Quantitative Risk Management: Concepts, Techniques and Tools, revised ed. Princeton University Press.
- Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). Nonstandard errors. *Journal of Finance*, 79(3), 2339–2390.
- Nelson, D.B. (1990). Stationarity and persistence in the GARCH(1,1) model. *Econometric Theory*, 6(3), 318–334.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Nolde, N., Ziegel, J.F. (2017). Elicitability and backtesting: perspectives for banking regulation. *Annals of Applied Statistics*, 11(4), 1833–1874.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

References (6/6)

- Patton, A.J., Ziegel, J.F., Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- Phillips, P.C.B., Shi, S., Yu, J. (2015). Testing for multiple bubbles: historical episodes of exuberance and collapse in the S&P 500. *International Economic Review*, 56(4), 1043–1078.
- Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). Specification curve analysis. *Nature Human Behaviour*, 4(11), 1208–1214.
- Stambaugh, R.F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375–421.
- Tsay, R.S. (2010). Analysis of Financial Time Series, 3rd ed. Wiley.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.
- Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). A tale of two time scales: determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411.